SE LAB1

Name: Aniket S Patil
SRN: PES2UG24CS064

Automated Rubric Assignment Evaluator - Requirements Table

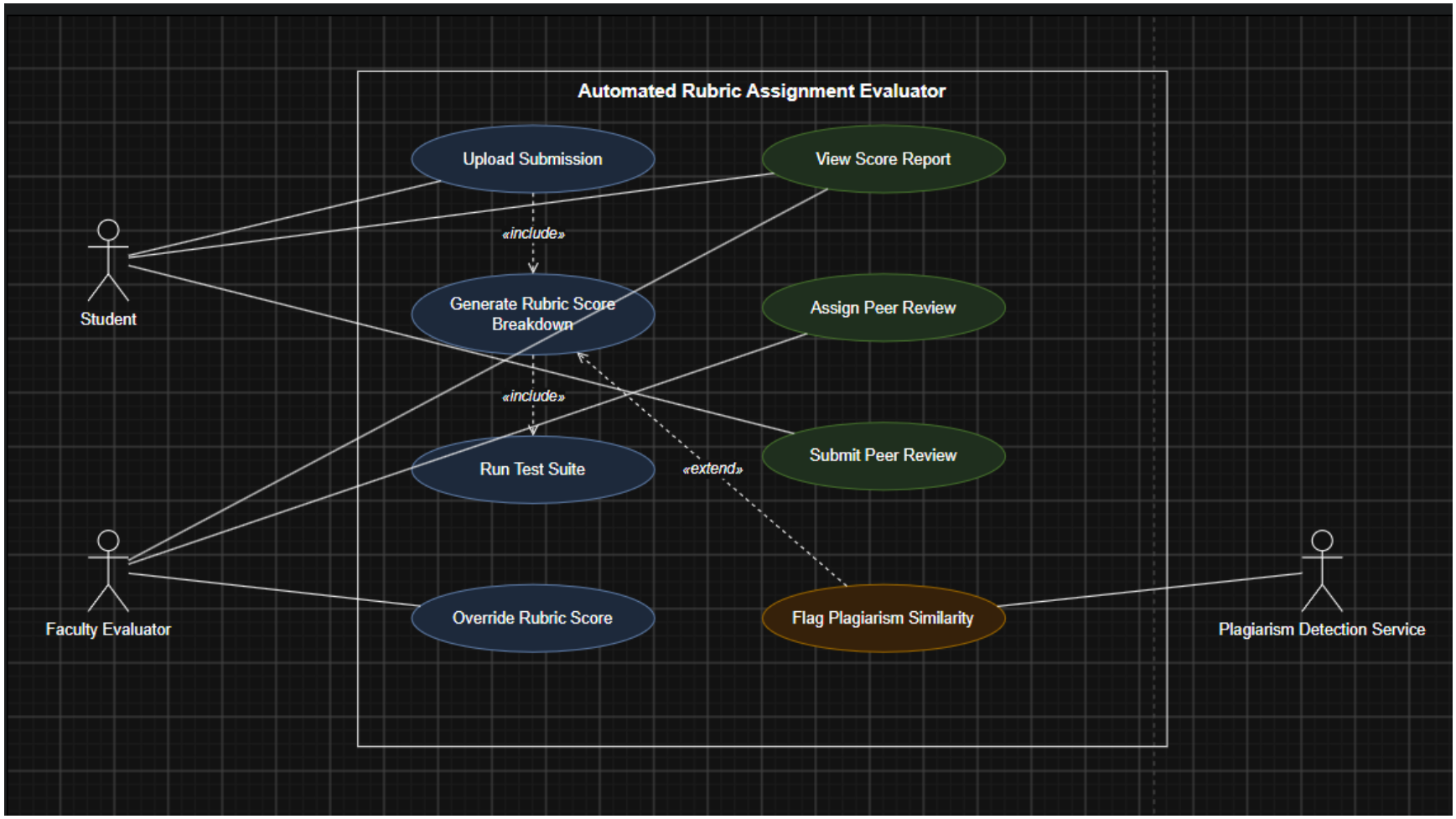
Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically queue uploaded student project zip files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	Pass: Rubric breakdown is rendered with test results within 10 seconds of submission. Fail: File queue stalls or invalid rubric tally.	Core function of the evaluator; removes manual grading overhead (given requirement).
FR-002	Functional	The system shall automatically assign each submission to two peer reviewers drawn from the same course section, excluding the submission's own author.	Medium	Pass: Every submission has exactly 2 distinct peer reviewers assigned within 5 minutes of the submission window closing. Fail: A reviewer is assigned to their own submission, or fewer than 2 reviewers are assigned.	Removes manual peer-review distribution overhead, as required by the problem context.
FR-003	Functional	The Faculty Evaluator shall be able to manually override any automatically generated rubric score, with a mandatory written justification.	High	Pass: An overridden score is saved together with the justification text and the item is flagged "manually reviewed." Fail: A score change is saved without justification text.	Automated scoring can misjudge edge cases (e.g., non-standard but correct code); faculty needs final authority.
FR-004	Functional	The system shall allow a Student to view their own itemized rubric score breakdown and export it as a PDF report.	Medium	Pass: A PDF report matching the displayed score is generated within 5 seconds of request. Fail: The report is missing rubric items or fails to generate.	Gives students transparency into how their grade was computed.
FR-005	Functional	The system shall run a similarity check across all submissions in the same batch and flag any pair exceeding a configurable similarity threshold.	High	Pass: Flagged pairs are listed with their similarity percentage, visible to the Faculty Evaluator. Fail: Submissions above the threshold are not flagged.	Protects academic integrity when grading is automated at batch scale.
NFR-001	Nonfunctional	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding 1.5 GB memory footprint.	High	Pass: Benchmarking tests confirm target latency and resource limits are met under simulated peak load.	Ensures the evaluator stays usable during high-traffic deadline periods (given requirement).
NFR-002	Nonfunctional	The system shall encrypt all stored submission files and rubric data at rest	High	Pass: An access-control audit confirms students cannot retrieve another	Protects academic records from tampering or leakage;

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		(AES-256) and enforce role-based access so students can only view their own scores.		student's rubric data via the API or UI. Fail: Unauthorized access to another student's data succeeds in testing.	student PII and grades are sensitive data.

Use-Case Flow Specification
Use Case: Generate Rubric Score Breakdown
System: Automated Rubric Assignment Evaluator
Primary Actor: System (triggered automatically as part of Upload Submission); results consumed by Student and Faculty Evaluator.
Preconditions: The Student has successfully uploaded a valid project zip file (Upload Submission is complete); the submission window for the assignment is open; a rubric definition and unit test suite are pre-configured for the assignment.
Postconditions: An itemized rubric score breakdown is generated and stored against the submission; the score is visible to the Student (via View Score Report) and the Faculty Evaluator; the submission's queue status is updated to "Evaluated" or "Pending Manual Review."
Main Success Scenario
1. Student uploads a project zip file (Upload Submission use case).
2. System validates the file format and adds the submission to the evaluation queue.
3. System retrieves the pre-configured unit test suite and rubric definition for the assignment.
4. System executes the test suite against the submitted code («include» Run Test Suite).
5. Test suite returns pass/fail results for each test case.
6. System maps the test results to rubric criteria and computes an itemized score for each criterion.
7. System checks the submission against the similarity threshold; if exceeded, Flag Plagiarism Similarity is triggered («extend»).
8. System stores the itemized rubric breakdown against the submission record.
9. System renders the rubric breakdown and notifies the Student and Faculty Evaluator within 10 seconds of submission.
10. Use case ends successfully.

Alternate Flow — 5a. Test Suite Execution Failure
5a1. System detects that the test runner has crashed or exceeded the configured execution timeout.
5a2. System marks the affected rubric criteria as "Unable to Evaluate" rather than assigning a score.
5a3. System queues the submission for manual review and notifies the Faculty Evaluator.
5a4. Faculty Evaluator manually enters scores for the flagged criteria (Override Rubric Score use case).
5a5. Use case resumes at main step 8 once manual scores are entered, or ends with submission status "Pending Manual Review" if not yet completed.


Automated Rubric Assignment Evaluator
Upload Submission
View Score Report
«include»
Student
Generate Rubric Score Breakdown
Assign Peer Review
«include»
Submit Peer Review
Run Test Suite
«extend»
Faculty Evaluator
Override Rubric Score
Flag Plagiarism Similarity
Plagiarism Detection Service